



Full length article

Does Pokémon Go lead to a more physically active life style?

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ABSTRACT

Previous research suggests that Pokémon Go can be employed as an intervention for promoting healthy behavior on a large scale. In contrast, some scholars expressed concerns about the long-term effects of this mobile game, indicating that the impact of the app on exercise is just moderate. We hypothesized that Pokémon Go users are more active, but this tendency is mainly driven by the greater physical activity needed to level up in the game and does not extend to physical activities in general. In the present study, 981 individuals completed a web survey, in which participants' frequency of Pokémon Go usage, overall physically active behavior, and amount of Pokémon Go related physical activity were measured. Regression analyses revealed that Pokémon Go related physical activity significantly reversed the positive effects of the app on participants' overall physically active behavior, suggesting that the mere adoption of the app does not reliably change people's behavior in general. The increase in physical activity levels is rather explained by the exercise required by the game.

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Pokémon Go makes everyone walk until our battery runs out.

–Anonymous, via Twitter.

1. Introduction

1.1. Theoretical background

Physical activity helps people maintain a healthy weight and reduces the risk for cardiovascular diseases, diabetes, obesity, high blood pressure, lipid disorders, and depression. Although this is widely recognized, many adults and children do not practice the recommended amounts of daily physical activity (World Health Organization, 2002). Previous research suggests that mobile technologies can spur people in becoming more active by providing an opportunity to engage in physically active behaviors (Fjeldsoe, Miller, O'Brien, & Alison, 2012; Kirwan, Duncan, Vandelandotte, & Mummary, 2012). More recently, a strategy for introducing physical activity into mobile technologies has been delineated by the development of mobile active video games (Barnett, Bangay, McKenzie, & Ridgers, 2013). They allow the game play to become

pervasive through continuous tracking (Arteaga, Kudeki, & Woodworth, 2010) and encourage multiple physical activities on an ongoing basis thanks to the introduction of new technologies such as augmented reality (Anderson et al., 2007), where game developers can combine virtual and real-world elements.

The first attempt in mixing virtual elements with real world physical activities was represented by the mobile game Ingress, in which virtual objects overlay a map of real geographical spaces via Google Maps. More recently, the mobile game Pokémon Go introduced great potential of mixing real-world activities with virtual worlds elements. Pokémon Go players use a mobile device's GPS to locate, capture, battle, and train virtual creatures, called Pokémon, appearing on the smartphone screen as if they were in the same real-world location as the player. The game uses a real-world map of streets and landmarks to help players locate Pokémon, Pokémon eggs (which will hatch once players have walked a specific distance, between 2 and 10 km), Pokéstops (for collecting special items), and Gyms (for challenging other players). The app has become a global phenomenon and was one of the best-selling apps in 2016, having been downloaded more than 650 million times worldwide and with 5 million daily active users globally (Smith, 2017). Due to its massive penetration and mix of virtual elements with real-world physical activities, Pokémon Go has been described as an intervention for promoting healthy behavior on a large scale (Althoff, White, & Horvitz, 2016).

Indeed, some studies suggested that Pokémon Go leads to

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significant increases in physical activity (Meschtscherjakov, Trösterer, Lupp, & Tscheligi, 2017), increasing users' steps by an average of 25% per day, across men and women of all ages, weight status and prior activity levels (Althoff et al., 2016). On the other hand, some scholars expressed concerns about the long-term effects of Pokémon Go on active behavior (Howe et al., 2016), indicating that the impact of Pokémon Go on exercise might be just moderate. They reported an increase in steps during the first week of installation and then a gradual decrease over the subsequent five weeks. By the sixth week, the number of daily steps had gone back to pre-installation levels (Howe et al., 2016). Given these mixed findings, we felt it an important endeavor to further examine the impact of Pokémon Go on physical activity. In particular, we were interested in whether this game can indeed trigger stable behavioral changes toward a more physically active life-style or whether any increases in physical activity reported in the existing literature are tied to the context of the game and do not translate into a stable behavioral change (Baranowski, 2016).

1.2. Research goal

It is worth noticing that Pokémon Go was not designed with the aim of increasing physical activity (see also Meschtscherjakov et al., 2017). Rather, moving around was a side-effect of playing the game successfully. Starting from these assumptions, the main goal of this study was to test whether the daily use of the Pokémon Go mobile app would increase people's physically active behaviors in general or whether the effects of the app were indeed limited to the physical activities needed to level up in the game (e.g., walk more to catch new Pokémon). To address this research question, in the present study, participants reported their frequency of Pokémon Go usage, the specific physical activity related to the use of Pokémon Go, and the level of their overall physically active behavior. Moreover, research on health-related interventions has pointed out that attitudes are one of the most important predictors of physical activity. Indeed, people who take benefits of physical activities usually have positive attitudes toward sports, whereas inactive individuals report negative beliefs about sports, leading to avoid exercise and its positive effects (Salehnia, Mizanj, Sajadi, & Rahimizadehd, 2012). Thus, in order to control for possible individual differences on beliefs about physical activity, we also assessed participants' propensity to engage in physically active behaviors in terms of attitudes toward physical activities.

We expected frequency of Pokémon Go usage to be positively associated with both specific physical activity related to the use of Pokémon Go as well as their overall physically active behavior. Crucially, we predicted that the frequency of Pokémon Go usage would no longer be positively associated with the overall physically active behavior when controlling for attitudes toward physical activity and for the impact of specific physical activity related to the use of Pokémon Go. In other words, we reasoned that Pokémon Go users are more active, but this tendency is completely driven by the greater physical activity related to the use of Pokémon Go and does not extend to general physical activities.

2. Method

The study was carried out in accordance with the code of ethics of the world medical association (Declaration of Helsinki) for experiments involving humans. Data collection was performed through Amazon Mechanical Turk (MTurk), an internet-based platform that provides an online participant pool (Buhrmester, Kwang, & Gosling, 2011; Paolacci & Chandler, 2014) and represents a method of rapid and inexpensive data collection (Johnson & Borden, 2012).

Participation in our study was voluntary and all participants were informed that responses were anonymous and no foreseeable risks were involved in participating in the study. Written informed consent was obtained from participants by requiring them to click on an "agree" button at the beginning of the survey. As a cover story, participants were told that the first part of the survey was related to their fitness and physical exercise habits, whereas the second part was composed of a few questions for a future study regarding the use of mobile games.

Since we had no previous research to rely on for estimating the size of our effects, we decided to run a large number of participants. Nine-hundred and ninety-nine individuals from the United States (617 females, 382 males; mean age = 32.51 years, $SD = 10.20$) participated in our study and received \$0.60. One potential drawback of Mechanical Turk experiments is that unsupervised participants tend to be less attentive than participants in a lab with an experimenter present (Oppenheimer, Meyvis, & Davidenko, 2009). Thus, to identify subjects who failed to pay close attention, we included one "catch-trial" in our online survey (i.e., "please answer 2 to this question"). Eighteen participants failed the attention check. Thus, the final sample consisted of 981 individuals (610 females, 371 males; mean age = 32.55 years, $SD = 10.16$).

After providing demographic data, participants' attitudes toward physical activity were assessed¹. We adapted the Attitude Regarding Physical Activities for Health and Fitness Scale (Khan, Abbass, Islam, Khan, & Din, 2012; six items; $\alpha = .68$). Sample items were "Participation in physical activities reduces the risk of heart diseases" and "To promote better health conditions, people may take part in the sporting activities", measured on a scale from 1 = "completely disagree" to 7 = "completely agree".

Participants' overall physically active behavior was assessed by measuring both recency and frequency of their physical activity during the month preceding the study ($\alpha = .77$). The six items adopted for measuring recency of participants' physical activity were "When was the last time you had (1) a walk for more than 30 min/(2) had a run/(3) had a bike ride to get some exercise?" on a scale anchored with 1 = "more than one month ago", 2 = "about four weeks ago", 3 = "about three weeks ago", 4 = "about two weeks ago", 5 = "about one week ago", 6 = "during the last week" and 7 = "yesterday". Frequency of past behavior was measured adopting the following items: "How many times have you had (1) a walk for more than 30 min/(2) had a run/(3) had a bike ride to get some exercise during the last month?" (anchored with 1 = "never", 2 = "two times", 3 = "from three to five times", 4 = "from six to eight times", 5 = "from nine to eleven times", 6 = "from twelve to fourteen times" and 7 = "every day").

Adopting one item, participants were then asked to report their frequency of Pokémon Go app usage. The item was "How often do you use the new Pokémon Go mobile app on your smartphone?" (on a scale from 1 = never to 7 = very often).

On an exploratory basis, we also assessed the extent to which people share their achievements on social media by adopting the following item "Do you usually share your Pokémon Go achievements on social networks with other friends in the Pokémon Go network?" (anchored with 1 = "never" to 7 = "very often").

Finally, the amount of specific physical activity related to the use of Pokémon Go was assessed adopting three items² ($\alpha = .73$). The items were "How many times have you walked more than 30 min/had a run/had a bike ride with the intent of searching for Pokémon Go during the last month?" anchored with 1 = "never", 2 = "two times", 3 = "from three to five times", 4 = "from six to eight times", 5 = "from nine to eleven times", 6 = "from twelve to fourteen times" and 7 = "every day".

3. Results

3.1. Preliminary analyses

Descriptive statistics (see Table 1) showed that on average participants had a positive attitude toward physical activity, and they reported being moderately physically active during the month preceding data collection. Moreover, 21% of participants reported to play Pokémon Go from *frequently* to *very often* (see Table 2). As expected, correlation analyses revealed that participants' overall physically active behavior was positively associated with frequency of app usage, attitudes toward physical activity, and Pokémon Go related physical activity, whereas attitudes were negatively associated with the frequency of Pokémon Go usage and the amount of Pokémon Go related physical activity. Moreover, social sharing was positively associated with both Pokémon Go app usage and Pokémon Go related physical activity.

We also examined whether there were gender differences for any of the dependent variables. No significant differences were found between males ($M = 2.52$, $SD = 2.07$) and females ($M = 2.40$, $SD = 2.10$) regarding the frequency of Pokémon Go app usage, $t(979) = 0.85$, $p = .391$. However, male players reported higher scores of Pokémon Go related physical activity ($M = 1.61$, $SD = 1.06$) compared to female players ($M = 1.41$, $SD = 0.87$), $t(979) = 3.10$, $p = .002$. Thus, participant gender was included in the following analyses.

3.2. Main analyses

A three-stage hierarchical multiple regression was conducted with participants' overall physically active behavior as the dependent variable (see Table 3 for regression statistics). Tests to see if the data met the assumption of collinearity indicated that multicollinearity was not a concern (Pokémon Go app usage, Tolerance = .45, VIF = 2.21; Pokémon Go related physical activity, Tolerance = .45, VIF = 2.22; Attitudes, Tolerance = .98, VIF = 1.02; Gender, Tolerance = .97, VIF = 1.03) and the data met the assumption of independent errors (Durbin-Watson value = 2.06). Frequency of app usage was entered at Stage 1 of the regression, Pokémon Go related physical activity at Stage 2, attitudes at Stage 3, and gender at Stage 4.

The hierarchical multiple regression (see Table 3) revealed that at Stage 1 Pokémon Go app usage significantly predicted participants' overall physically active behavior and accounted for 1% of the variation in the dependent variable. Adding the amount of Pokémon Go related physical activity at Stage 2 explained an additional 9.8% of variation (see Table 3) and this change in R^2 was significant. As can be seen, frequency of Pokémon Go app usage still significantly predicted participants' overall physically active behavior. However, the effect was no longer positive but negative. That is, the introduction of Pokémon Go related physical activity in the regression model reversed the positive effects of the app on participants' overall physically active behavior, suggesting that

Table 2

Distribution of participants' Pokémon Go app usage.

	N	%
Never	588	59.9
Rarely	67	6.8
Occasionally	46	4.7
Sometimes	74	7.5
Frequently	73	7.4
Often	49	5.0
Very Often	84	8.6
Total	981	100%

Pokémon Go related physical activity serves as a suppressor. The effect of frequency of Pokémon Go app usage remained significant when controlling for attitudes and for participant gender, indicating that the negative association between frequency of Pokémon Go usage and participants' overall physical activity is independent from existing beliefs about physical activity and gender.

It thus appears that the positive effect of Pokémon Go usage on activity is limited to activity related to the app but does not generalize to other activities. Interestingly, attitudes toward physical activity were negatively correlated with both frequency of app usage and Pokémon Go related physical activity (Table 1), which suggests that using the app is not associated with a more favorable opinion toward exercise in general. Overall, we suspect that Pokémon Go users will be less active when they no longer use the app.

4. Discussion

The Pokémon Go phenomenon has reached millions of people since its release in 2016, and some scholars have highlighted potential benefits such as increased physical activity, spending more time outside, and fostering social interactions (Zach & Tussyadiah, 2017). Yet, other researchers reported concerns about the duration of those effects (Chaput & LeBlanc, 2017; Howe et al., 2016). Starting from these assumptions, the present research tested and confirmed the hypothesis that the positive effect of Pokémon Go in promoting a physically active life-style is limited to the physical activities needed to level up in the game but does not extend to physically active behavior in general.

Although we cannot establish any causal direction of the effect, the negative correlation of attitudes with both frequency of app usage and Pokémon Go related physical activity suggests that the use of Pokémon Go does not lead people to be more favorable toward exercise in general. Indeed, one recent systematic review on the effects of active video games as a tool for promoting physical activity suggested that it is unclear to date whether active video games can be used effectively in the long term to reliably increase daily physical activity (Biddiss & Irwin, 2010; Chaput & LeBlanc, 2017). Furthermore, the introduction of Pokémon Go related physical activity in the model significantly eliminated the positive effects of the app on participants' overall physically active behavior

Table 1

Means, standard deviations, and bivariate correlations.

	M	SD	1	2	3	4	5
1. Frequency of Pokémon Go app usage	2.45	2.08					
2. Pokémon Go related physical activity	1.49	0.95	.74**				
3. Overall physically active behavior	3.24	1.35	.01**	.28**			
4. Attitudes toward physical activity	5.87	0.70	-.07*	-.10**	.07*		
5. Social sharing	1.61	1.40	.68**	.66**	.10**	-.08**	
6. Participant gender	—	—	.03	.10**	.12**	-.12**	-.00

* $p < .05$, ** $p < .01$. Gender coded 1 = female and 2 = male.

Table 3

Hierarchical regression analysis assessing Pokémon Go app usage, Pokémon Go related physical activity, attitudes, and gender as predictors of participants' overall physically active behavior.

Predictor	Step 1			Step 2			Step 3			Step 4		
	B	SE	β	B	SE	β	B	SE	β	B	SE	β
Frequency of Pokémon Go app usage	.064	.021	.098***	-.161	.029	-.248***	-.162	.029	-.249***	-.156	.029	-.239***
Pokémon Go related physical activity				.665	.064	.467***	.679	.064	.477***	.657	.064	.461***
Attitudes							.185	.058	.096***	.204	.058	.106***
Gender										.262	.085	.093***
R^2	.009***			.106***			.114***			.122***		
F for change in R^2	9.48**			107.78***			10.00**			9.49**		

Note. *** $p < .001$, ** $p < .01$.

(in fact, it reversed it). This pattern confirms previous literature showing that Pokémon Go allows players to increase the recommended daily volume of physical activity (Althoff et al., 2016) by involving them in exercise necessary to progress in the game. However, the negative effect of Pokémon Go on people's overall physically active behavior suggests that the mere adoption of the app does not reliably change people's general behavior.

These findings are consistent with previous research on video game exposure showing that playing video games can contribute to a sedentary lifestyle (Baranowski et al., 2012; Vandewater, Shim, & Caplovitz, 2004). We surmise that the mere adoption of the Pokémon Go app does not yield stable changes in people's overall physically active behavior. In fact, individuals can use the app without engaging in extra exercise and instead limit their activities to discover new Pokémon when moving around for different, unrelated reasons. Indeed, even if the app adopted a speed limit control of 10.5 km per hour, it is common practice among users to activate Pokémon Go when travelling by car (Ayers et al., 2016). It could be that those playing more frequently also fall into this user type, that is, looking for new Pokémon when doing something else in a new and unexplored place. Thus, a second interpretation is that frequent users might play Pokémon Go more intermittently, being physically active only some of the times they play the game. Finally, a third interpretation might be that users who have already gone out for a walk looking for Pokémon simply do not want to engage in additional sport activities.

Taken together, we can only speculate on why using the app is not positively associated with attitudes towards and engagement in physical activity. Prior research suggested that the main factors of success in health and wellbeing interventions are positive reinforcement and rewards (see Johnson et al., 2016). Indeed, the elements of Pokémon Go that are most successful in pushing users to be more physically active are those that directly relate to the ability to catch new Pokémon (Meschtscherjakov et al., 2017). Pokémon Go does reward players with stardust and coins, yet such rewards are not usually gained by engaging in physical activity. Thus, with more contingency between rewards and app usage that requires physical activity, the introduction of specific rewards for triggering positive reinforcement in Pokémon Go may have positive effects on attitudes and behavior related to sports and health.

5. Limitations and directions for future research

Some limitations of the present study must be acknowledged. First, participants' physically active behavior and Pokémon Go related physical activity were self-reported and activity levels might be partly driven by social desirability. As expected, the frequency of app usage and Pokémon Go related physical activity were strongly correlated, so the change in sign of the regression model may be due to statistical artefacts which result from using inter-correlated predictors. Indeed, when the predictors are correlated,

the effect of one predictor may be exaggerated and the other is driven in the opposite direction to compensate, sometimes producing a spurious effect that reaches significance in the opposite direction (Maassen & Bakker, 2001). Nevertheless, we do not think that this is the case, since the theoretical background provided by prior literature on the effects of active video games such as Pokémon Go support our results (Baranowski et al., 2012; Howe et al., 2016; Vandewater et al., 2004). Yet, to better understand the relationship between physical activity and specific physical activity due to the use of an active mobile video game, future studies should measure participants' actual behavior adopting distance measuring applications or pedometers to obtain more precise and reliable data.

Furthermore, there are many other psychological factors which affect and sustain behavioral change (see, for example, Fogg, 2009; Schwarzer, 2008). However, only attitudes were considered in our study. Thus, future research should consider the role of other variables such as self-efficacy, perceived behavioral control, and individual motivation. In this regard, research on social media has highlighted that social sharing of tracked results can increase motivation (Welmo & Neville, 2017) and foster positive behavioral reinforcement (Allison, 2012). Correlation analyses suggested that social sharing was positively associated with both frequency of Pokémon Go usage and Pokémon Go related physical activity. However, Pokémon Go does not natively provide users with social sharing features and players have to manually take screenshots of the game and share them on their personal social media profiles. Thus, we speculate that adding social features to the game (e.g., sharing the daily number of steps made to catch Pokémon) may provide incentives for engaging in the physical activities proposed by the app.

We also considered a short period of time for our study, and long-term impact of mobile active video game such as Pokémon Go on physical activity levels can also be further explored adopting a longitudinal design. Furthermore, we did not differentiate participants in terms of game achievements (level, Pokédex completion, etc.), and we just considered the frequency of play as a proxy for intensity of play. Future studies may differentiate between novel and experienced users.

6. Conclusion and outlook

Our results suggest that players are more physically active when they play Pokémon Go, but these effects do not reliably spur people toward a more physically active life-style in general. Previous studies proposed Pokémon Go as an instrument for health promotion interventions on a large scale (e.g., Althoff et al., 2016). It indeed appears that active mobile games such as Pokémon Go represent a useful strategy for promoting physical exercise as long as they are played regularly by people over time. However, it seems that as soon as individuals no longer use the app, the positive

impact on activity vanishes.

In general, augmented reality technology shows potential for promoting health programs (Biddiss & Irwin, 2010; Chaput & LeBlanc, 2017). Yet, one of the issues of using computer and mobile games to promote behavioral change is that game involvement usually decreases with time (Hswen, Murti, Vormawor, Bhattacharjee, & Naslund, 2013). Indeed, as proposed by other scholars (Gao, 2017), augmented reality video games need to be constantly developed to hold people's interest and promote stable behavioral changes.

Indeed, smartphones, being pervasive devices, are capable of keeping user's motivation longer than conventional approaches (Maitland et al., 2006). Thus, mobile app designers should consider embedding appropriate feedback cues highlighting the benefits of physical activities so to gradually shift players' focus outside the game, toward the adoption of a physically active life style in general, even when people quit playing the game.

7. Notes

- Our survey included another scale for measuring participants' attitudes. The measure adopted a 7-point semantic differential's adjectives measure ($\alpha = .75$). The items were introduced by the sentence "For each pair of adjectives, please select the value between them which reflects the extent to which you believe the adjectives describe sports activities. I think that engaging in physical active behaviors, is foolish/wise, pleasant/unpleasant (reverse scored), bad/good, depressing/funny, unhealthy/healthy. Scales were anchored by 1 assigned to the negative pole and 7 to the positive one, with 4 indicating neither/nor. Running an alternative regression analysis by considering this measure confirmed our findings. Pokémon Go app usage was negatively associated with participants' overall active behavior ($b = -.148$, $SE = .029$, $\beta = -.228$, $t(977) = 5.18$, $p < .001$) when controlling for Pokémon Go related physical activity ($b = .667$, $SE = .062$, $\beta = .469$, $t(977) = 10.70$, $p < .001$) and attitudes ($b = .359$, $SE = .049$, $\beta = .218$, $t(977) = 7.39$, $p < .001$; $R^2 = .15$, $F(3, 977) = 59.79$, $p < .001$).
- We also assessed a fourth item for measuring the Pokémon Go related physical activity variable. The item was "How often do you go out searching for Pokémon?", measured on a scale 1 = "never" to 7 = "very often". Combining this item with the three items described in the method section resulted in collinearity problems (Pokémon Go app usage, Tolerance = .22, VIF = 4.37; Pokémon Go related physical activity, Tolerance = .22, VIF = 4.37; Attitudes, Tolerance = .99, VIF = 1.01). Indeed, there is concern when VIF is greater than 2.5 and Tolerance is lower than .4036. Because this item was not consistent with the other three, it was dropped from the main analyses.

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